Notes: Copeland, Martin, and Walker (JF, 2014)

Repo Runs: Evidence from the Tri-Party Repo Market

Contents

1 Big picture

- **Question.** Did the U.S. repo market experience a system-wide run during the 2007–2009 financial crisis?
- **Motivation.** Repo was central to crisis narratives because dealers relied heavily on short-term collateralized funding, and earlier work documented sharply rising margins in part of the bilateral repo market.
- **Main answer.** The paper argues that there was **no system-wide run in the tri-party repo market**. Margins and funding were surprisingly stable for most dealers and most collateral classes.
- **Important exception.** Lehman Brothers experienced a sharp collapse in tri-party repo funding in September 2008, but the collapse was concentrated and did not look like a broad run on all dealers.
- **Key distinction.** The bilateral repo market tightened significantly, especially dealer-to-client repo, but the tri-party repo market behaved differently. The paper shows that repo market fragility cannot be inferred from one segment alone.
- **Core empirical facts.** Government collateral margins were almost flat; nongovernment collateral margins rose only modestly; collateral volumes in tri-party repo declined gradually; and daily dealer-investor relationships were stable.
- **Mechanism emphasized.** In tri-party repo, cash investors appear more likely to reduce or withdraw funding than to demand sharply higher margins. This differs from the classic margin-spiral view.
- **Policy relevance.** The findings complicate the case for uniform minimum margins across repo markets. Margin policy may matter more in bilateral repo than in tri-party repo.
- **Takeaway.** The crisis involved severe stress in short-term funding markets, but tri-party repo did not suffer a generalized margin run. Its fragility was more about concentrated dealer failure, collateral liquidation, and institutional plumbing than about broad margin spirals.

2 Incremental contribution of the paper

2.1 Contribution relative to Gorton and Metrick-style repo run evidence

- The paper directly responds to the view that repo markets experienced a broad run through sharply increasing haircuts or margins.
- Earlier evidence focused on segments of bilateral repo, especially transactions among high-quality dealers or dealer-to-client relationships.
- Copeland, Martin, and Walker bring confidential data from the tri-party repo market, a major repo segment that was central to dealer funding.
- The main incremental contribution is to show that the crisis pattern in bilateral repo does not generalize to tri-party repo.

Interpretation: The paper does not deny that parts of repo markets became stressed. Its contribution is segment-level precision: repo markets are not homogeneous, and run dynamics differ across segments.

2.2 Contribution to the theory of runs and margins

- Standard run narratives often emphasize rising haircuts, margin spirals, and collateral liquidation pressure.
- The paper shows that in tri-party repo, margins were mostly stable even during severe crisis months.
- This suggests that cash investors may respond to dealer stress by pulling funding rather than continuously repricing collateral through higher margins.

- The Lehman episode illustrates a nonlinear funding collapse: collateral posted by Lehman fell abruptly while margins moved little.

Interpretation: The incremental theoretical lesson is that a run can occur through a quantity channel rather than a margin channel. A dealer may lose funding without observing large margin increases.

2.3 Contribution to policy debates on minimum margins

- The paper was written during policy debates over minimum repo margins and macroprudential haircut regulation.
- If margins in a market are highly procyclical, minimum margins may reduce destabilizing margin spirals.
- The paper finds little procyclicality in tri-party margins, so the benefits of minimum margins may be smaller in this segment.
- The evidence suggests that policy should distinguish bilateral repo, tri-party repo, and GCF repo rather than treating all repo as one market.

Interpretation: The paper's incremental policy contribution is not to reject repo regulation. It is to identify where the relevant fragility lies and which policy instruments are likely to matter.

2.4 Contribution to empirical measurement of shadow banking

- The paper uses supervisory/confidential tri-party repo data, which were rarely available in academic research at the time.
- It documents collateral composition, margins, dealer-investor relationships, and Lehman-specific dynamics.
- It connects these market microstructure facts to crisis narratives and policy debates.

Interpretation: The paper is partly a measurement contribution. It replaces broad conjectures about repo funding with detailed evidence from a central market segment.

3 Market structure and data

3.1 Repo contracts and margin

A repo is economically similar to a collateralized loan. A cash investor lends cash to a dealer or borrower and receives securities as collateral. The dealer promises to repurchase the securities at a future date.

The repo margin is

$$Margin = \frac{Collateral\ Value - Cash\ Lent}{Cash\ Lent}. \tag{1}$$

- A higher margin means the cash lender is more overcollateralized.
- A margin increase tightens funding conditions for the dealer because the dealer must post more collateral per dollar of funding.
- In a margin-spiral story, margins rise when perceived risk increases, forcing deleveraging and asset sales.

Interpretation: Margin is the key object in a haircut-run narrative. Stable margins therefore provide direct evidence against a broad haircut run in tri-party repo.

3.2 Tri-party repo versus bilateral repo

- In bilateral repo, the two counterparties manage settlement and collateral directly.
- In tri-party repo, a clearing bank provides settlement and collateral management services.
- Tri-party cash investors include money market mutual funds, securities lenders, and other institutional investors.
- Tri-party collateral providers are securities dealers.
- The tri-party market is mostly overnight and is central to dealer short-term funding.

Interpretation: The clearing-bank infrastructure and investor base make tri-party repo different from bilateral repo. This helps explain why margins behaved differently across the two markets.

3.3 Main tri-party repo data

- The data cover daily tri-party repo transactions from July 2008 to January 2010.
- The authors observe collateral class, margin, dealer, investor, and dollar amount.
- The sample covers the crisis period around Lehman Brothers' bankruptcy and the post-crisis stabilization period.
- The paper divides the sample into a **crisis period** from July 2008 to July 2009 and a **stable period** from July 2009 to January 2010.

Interpretation: Daily transaction-level data let the authors examine whether margins jump, whether volumes collapse, and whether dealer-investor relationships break down.

3.4 Collateral classes

The paper divides collateral into government and nongovernment groups.

- **Government collateral:** U.S. Treasuries and strips, agency debentures, agency MBS, and agency REMICs.
- **Nongovernment collateral:** asset-backed securities, commercial paper, corporate bonds, equities, municipal bonds, private-label CMOs, whole loans, and residual other collateral.

Interpretation: The distinction matters because a run is more likely to show up in lower-quality or less liquid collateral classes. The paper finds the largest issues in nongovernment collateral, not in government collateral.

3.5 Lehman Brothers data and stress measures

- Lehman is analyzed separately because it filed for bankruptcy on September 15, 2008.
- The paper tracks Lehman's margins, collateral composition, number of investors, and collateral posted in the days leading to bankruptcy.
- The authors compare Lehman's margin to the average margin faced by other large dealers.
- The margin spread is Lehman's average margin minus the average margin of other large dealers.

Interpretation: Lehman is the key case study because it is where a true run-like collapse appears in the tri-party repo data.

4 Empirical design and specifications

4.1 Descriptive comparison of margins and quantities

The first empirical strategy is descriptive. The paper asks:

1. Do margins increase sharply during the crisis?
2. Do collateral quantities collapse broadly across dealers?
3. Do changes differ by collateral type?
4. Are changes concentrated in Lehman Brothers?

Interpretation: The paper's main evidence comes from direct measurement rather than a single regression. This is appropriate because the central claim is about whether tri-party repo margins and funding moved in run-like ways.

4.2 Daily collateral-change distributions

The paper computes daily percentage changes in collateral posted by dealers and received by investors:

$$\Delta Collateral_{i,t} = 100 \times \frac{Collateral_{i,t} - Collateral_{i,t-1}}{Collateral_{i,t-1}}. \tag{2}$$

It then reports the 10th, 25th, 50th, 75th, and 90th percentiles of these changes.

Interpretation: If investors or dealers frequently broke relationships, these distributions would show large day-to-day movements. The median near zero indicates stable rollover relationships.

4.3 Margin decomposition regressions

In the appendix, the paper estimates regressions of the form

$$Margin_{djt} = \alpha_0 + \sum_d \eta_d Dealer_d + \sum_j \zeta_j CollateralType_j + \alpha_1 ClearingBank_t + \varepsilon_{djt}. \tag{3}$$

- $Margin_{djt}$ is the margin for dealer d, collateral type j, and date t.
- Dealer fixed effects capture persistent differences in dealer margin levels.
- Collateral-type fixed effects capture margin differences across asset classes.
- Clearing-bank controls account for settlement-bank differences.

Interpretation: The decomposition shows that both dealer identity and collateral type matter for margins. Dealer fixed effects are large, indicating persistent cross-dealer heterogeneity.

4.4 CDS regressions

The paper relates dealer credit risk to margins and collateral quantities using regressions of margins and collateral value on CDS spreads and squared CDS spreads:

$$Y_{dt} = \alpha + \beta_1 CDS_{dt} + \beta_2 CDS_{dt}^2 + \varepsilon_{dt}, \tag{4}$$

where Y_{dt} is either margin or collateral value, separately for government and nongovernment collateral.

Interpretation: If cash investors in tri-party repo react smoothly to credit risk, dealer CDS spreads should explain margins and collateral quantities. The paper finds little evidence that they do.

5 Identification and interpretation

5.1 What the evidence identifies

- The paper identifies how one major repo segment actually behaved during the crisis.
- It shows whether the tri-party repo market as a whole exhibited rising margins and broad funding withdrawals.
- It distinguishes system-wide tri-party repo stress from idiosyncratic dealer failure.

Interpretation: The paper does not identify all repo-market behavior. It identifies behavior in tri-party repo and contrasts it with documented bilateral repo patterns.

5.2 Why tri-party and bilateral repo differ

- Different investor bases: tri-party repo involves money funds and securities lenders, while bilateral repo includes dealers and hedge funds.
- Different collateral management: tri-party uses clearing banks; bilateral transactions are managed directly by counterparties.
- Different role of rehypothecation and client financing.
- Different responses to stress: bilateral lenders may raise margins, while tri-party cash investors may withdraw cash or refuse rollover.

Interpretation: The same shock can produce margin increases in one segment and quantity withdrawals in another.

5.3 Limits of the paper's evidence

- The paper does not observe the full bilateral repo market directly; it relies on other sources and limited confidential data for comparison.
- The tri-party data begin in July 2008, so they do not capture the earliest phase of the crisis in 2007.
- The data show stable margins, but stable margins do not imply absence of all fragility.
- Lehman demonstrates that tri-party repo can still be fragile at the dealer level.

Interpretation: The paper's conclusion is precise: no system-wide tri-party repo margin run is observed. It is not a claim that repo markets were safe or that short-term funding was unimportant.

6 Tables and figures

6.1 Figure 1 and Figure 2: margins in tri-party repo versus bilateral repo

Read:

- Figure 1 shows that median tri-party repo margins were stable for government collateral and rose only modestly for nongovernment collateral.
- Treasury and agency-related margins remain close to 2% throughout the crisis.
- Nongovernment margins are higher and rise more, but the increase is far smaller than in bilateral repo.
- Figure 2 compares median margins across bilateral and tri-party repo by asset class.
- The spread between bilateral and tri-party margins is large for some asset classes, especially ABS and private-label CMO collateral.

Economic message: The main figure pair shows that the run-like rise in margins documented in bilateral repo is not present in the same way in tri-party repo.

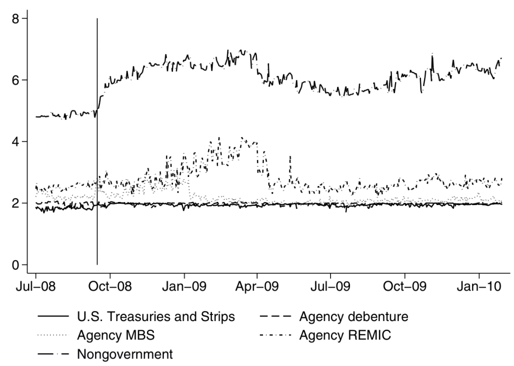

Figure 1. Median margins in tri-party repo by asset class. This figure presents the dai time series of the median margin by asset class in tri-party repo. Margin is equal to the ratio the value of collateral posted over the amount of cash lent minus one and is reported as a percer The vertical line denotes the date of Lehman Brothers' bankruptcy filing. MBS is mortgage-back securities and REMIC is real estate mortgage investment conduits. Nongovernment securiti are those not backed by the full faith and credit of the federal government. Examples are equitie corporate bonds, and other securities not issued by the U.S. Treasury or other government agencie

Figure 1: median tri-party repo margins by asset class

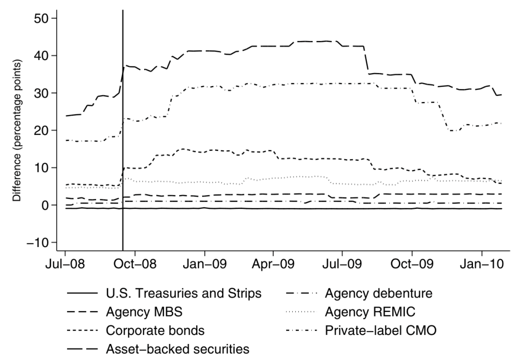

Figure 2. Differences in median margins in tri-party repo by asset type. This figur presents the daily time series of the difference in median margins for an asset class across th bilateral and tri-party repo markets. Differences in median margins are calculated as the bilatera repo margin median minus the tri-party repo margin median for each asset class. The vertical blacl line corresponds to Lehman Brothers' bankruptcy. MBS is mortgage-backed securities, REMIC i real estate mortgage investment conduits, and CMO is collateralized mortgage obligations.

should not expect to see higher margins in the tri-party repo market. Rather

Figure 2: bilateral minus tri-party margin differences

6.2 Figure 3 and Table I: market structure and collateral composition

Read:

- Figure 3 maps the main segments of the U.S. repo market and places tri-party repo in relation to bilateral repo and GCF repo.
- Securities dealers stand at the center of the market and borrow from cash investors against collateral.
- Table I shows that government collateral dominates the tri-party repo market.
- Government collateral accounts for about 82% of collateral, with agency MBS and Treasury collateral as the largest groups.
- Nongovernment collateral is about 18%, including corporate bonds, equities, ABS, and private-label CMO.

Economic message: The market was mainly backed by relatively safe government-related securities, which helps explain why broad margin increases were limited.

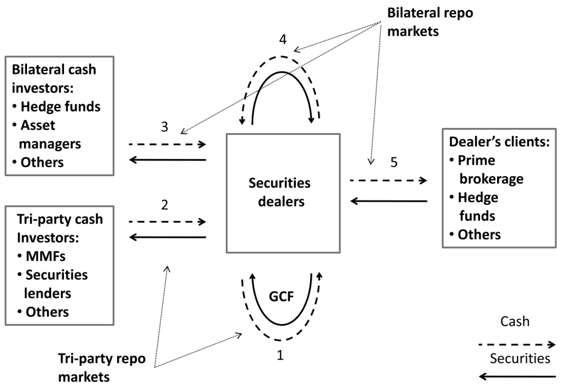

Figure 3. Stylized map of the U.S. repo market. This figure presents a map of the various U. repo markets. MMF is money market mutual funds and GCF is the General Collateral Finan Repo Service operated by the Fixed Income Clearing Corporation. This figure was taken from post cowritten by this paper's authors on the Liberty Street Economics site (see the blog post

Figure 3: stylized map of U.S. repo market

Table I

Composition of Tri-Party Repo Collateral

This table presents the composition of securities posted as collateral in tri-party repo as a percent of the total, over the entire sample as well as two sample periods. "Crisis" is the period from July 2008 to July 2009, "Stable" is from July 2009 to January 2010, and "All" covers both sample periods. MBS is mortgage-backed securities and REMIC is real estate mortgage investment conduits. Government securities are those backed by the full faith and credit of the federal government.

		Crisis	Stable	All
	Agency Debentures	12.6	11.4	12.2
Government	Agency MBS	38.3	37.5	38.0
Collateral	Agency REMIC	4.3	4.8	4.5
	U.S. Treasuries and Strips	26.8	29.0	27.4
	subtotal	82.1	82.7	82.2
	Asset-Backed Securities	2.2	2.4	2.2
	Commercial Paper	0.4	0.3	0.4
Nongovernment	Corporate Bonds	5.5	5.9	5.6
Collateral	Equities	4.1	4.0	4.1
	Money Market	1.3	1.6	1.4
	Municipal Bonds	0.9	0.7	0.9
	Other[a]	0.2	0.4	0.3
	Private-Label CMO	2.7	2.0	2.5
	Whole Loans	0.7	0.1	0.5
	subtotal	18.0	17.5	17.9

[a]The "Other" collateral class is a residual category and so the composition of its collateral changes over time.

Table I: collateral composition

6.3 Figure 4 and Figure 5: market size and lower-quality collateral

Read:

- Figure 4 shows that the total tri-party repo market declined from above \$2.5 trillion to around \$1.5 trillion over the sample.
- The decline was gradual rather than a sudden system-wide break.
- Figure 5 shows that corporate bonds and ABS collateral declined substantially after mid-2008.
- ABS fell sharply, and corporate bond collateral also declined, consistent with stress in lower-quality collateral.

Economic message: Quantities did decline, especially for lower-quality collateral, but the pattern is gradual and collateral-specific rather than a uniform market-wide run.

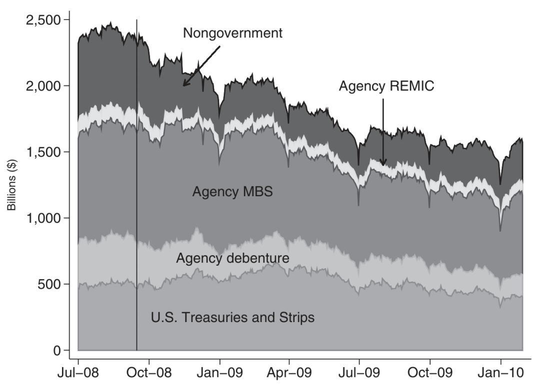

Figure 4. Tri-party repo market size. This figure describes the composition of collateral poste in tri-party repo over time. The vertical line represents Lehman Brothers' bankruptcy. MBS i mortgage-backed securities and REMIC is real estate mortgage investment conduit. Nongoverr ment securities are those not backed by the full faith and credit of the federal government. Exam ples are equities, corporate bonds, and other securities not issued by the U.S. Treasury or othe federal government agencies.

Across these two periods, the composition of collateral posted in tri-part repo did not change substantially (see Table I). We categorize all the securitie

Figure 4: tri-party repo market size

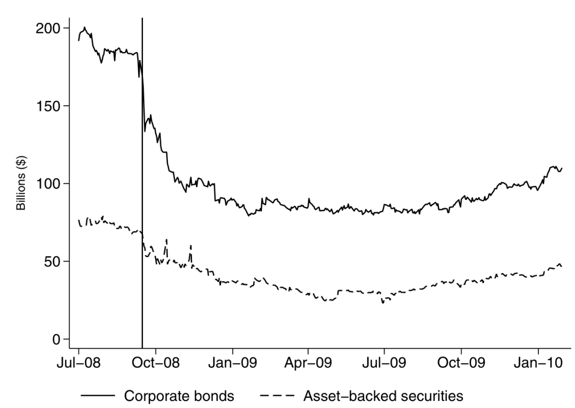

Figure 5. Use of corporate bonds and asset-backed securities as collateral in the party repo market. This figure presents daily time series of the total value of corporate b and asset-backed securities used as collateral in tri-party repo. The vertical line represents Leh Brothers' bankruptcy.

Figure 5: ABS and corporate bond collateral

6.4 Table II and Table III: margins and asset-class matching

Read:

- Table II shows that average margins are low for government collateral and higher for nongovernment collateral.
- U.S. Treasuries and strips have average margins around 1.7%.
- Equities, municipal bonds, private-label CMOs, ABS, and whole loans have higher margins.
- The average margin for all collateral is 2.8%, with little difference between crisis and stable periods.
- Table III maps collateral classes in bilateral repo to the closest tri-party collateral classes.

Economic message: The margin hierarchy looks sensible by collateral quality, but margins do not jump in crisis in the way a generalized margin run would predict.

Table II

Tri-Party Repo Margins: Mean and Standard Deviation

This table presents statistics on the margins observed in tri-party repo. Margin corresponds to the amount a repo transaction is overcollateralized and is equal to the ratio of the value of collateral posted over the amount of cash lent minus one (reported as a percent). "Crisis" margins are computed over July 2008 to July 2009, "Stable" margins over July 2009 to January 2010, and "All" margins over the entire sample period; "*SD*" is standard deviation. Securities are ordered from high to low quality, based on an ordering obtained from conversations with market participants. MBS is mortgage-backed securities, REMIC is real estate mortgage investment conduits, and CMO is collateralized mortgage obligations. Government securities are those backed by the full faith and credit of the federal government.

		Crisis		Stable		All	
		Mean (%)	*SD*	Mean (%)	*SD*	Mean (%)	*SD*
	U.S. Treasuries and Strips	1.7	0.59	1.8	0.42	1.7	0.54
Government	Agency Debentures	1.9	0.49	1.9	0.39	1.9	0.47
Collateral	Agency MBS	2.3	0.59	2.0	0.40	2.2	0.56
	Agency REMIC	3.1	1.31	2.6	0.59	2.9	1.15
	Money Market	3.8	1.29	4.1	1.19	3.9	1.26
	Commercial Paper	4.2	1.75	3.9	0.63	4.1	1.57
	Corporate Bonds	6.2	2.80	6.0	1.71	6.1	2.50
Nongovernment	Equities	6.3	1.57	8.5	2.28	7.0	2.08
Collateral	Municipal Bonds	7.7	7.74	5.3	3.76	7.1	7.04
	Private-Label CMO	6.3	2.83	5.9	3.43	6.2	2.99
	Asset-Backed Securities	7.1	3.90	5.8	1.73	6.7	3.40
	Whole Loans	8.7	1.16	8.3	4.74	8.7	1.58
	Other[a]	4.4	6.30	3.4	1.29	3.9	4.66
All		2.8	2.22	2.7	2.00	2.8	2.16

[a]The "Other" collateral class is a residual category and so the composition of its collateral changes over time. Consequently, the statistics on the mean and standard deviation of this collateral class are hard to interpret. Nevertheless, we include these statistics for completeness.

Table II: tri-party repo margin means and standard deviations

Table III

Matching of Asset Classes

This table presents how we match asset classes in the bilateral and tri-party repo data. MBS is mortgage-backed securities, REMIC is real estate mortgage investment conduits, and CMO is collateralized mortgage obligations.

Dealers as Cash Providers (Bilateral)	Dealers as Collateral Providers (Tri-Party)
U.S. Treasuries	U.S. Treasuries and Strips
Agency Debentures	Agency Debentures
Agency MBS	Agency MBS
Agency CMO	Agency REMIC
High-Grade Corporate Debt	Corporate Bonds
Alt-A, Prime MBS	Private-Label CMO
Subprime	Asset-Backed Securities

securities lenders. In contrast, our estimated coefficients on the dealer fixed

Table III: matching of bilateral and tri-party asset classes

6.5 Table IV and Table V: stability of dealer-investor relationships

Read:

- Table IV reports percentiles of daily changes in collateral posted by dealers.
- For small and large dealers, the median daily change is essentially zero.
- The tails of the distribution show that collateral can move materially, but most days exhibit stable positions.
- Table V reports the same concept for investors receiving collateral.
- Investor-level changes are also centered near zero across size quartiles.

Economic message: The stable day-to-day positions indicate that tri-party repo relationships mostly rolled over during the crisis rather than constantly breaking.

Table IV

Distribution of the Daily Change in the Value of Collateral Posted by Dealers

This table presents the percentiles of the distribution of the change in collateral posted by dealer type. An observation is the daily percent change in the total value of collateral posted by a dealer for a given collateral class. Quarter-end dates and all observations on Lehman Brothers are excluded from this analysis. For small and large dealers, there are 30,053 and 51,272 observations, respectively.

	Percentiles				
Type of Dealer	10th	25th	50th	75th	90th
Small	−24.3	−2.6	0.0	1.9	18.3
Large	−14.8	−3.5	−0.0	3.0	12.0

Table IV: daily collateral changes by dealers

Table V

Distribution of the Daily Change in the Value of Collateral Received by Investors

	Percentiles				
Investor Size (Quartiles)	10th	25th	50th	75th	90t
1	−5.1	−0.0	0.0	0.0	4.
2	−26.5	−2.8	0.0	2.1	20.
3	−15.7	−2.3	0.0	1.8	13.
4	−10.2	−2.5	0.0	2.2	8.

value of collateral received. Because there is substantial heterogeneity amon investors, we report these percentiles conditional on the size of the investo

Table V: daily collateral changes by investors

6.6 Table VI and Figure 6: Lehman Brothers as the central exception

Read:

- Table VI shows that Lehman's margin spread over other large dealers stayed small before bankruptcy.
- Lehman's share of nongovernment collateral increased in the days before failure.
- The number of Lehman investors fell only gradually until very late in the episode.
- Figure 6 shows a sharp collapse in the total collateral Lehman posted in tri-party repo around September 12–15, 2008.
- The collapse is concentrated and nonlinear, unlike the gradual broader market decline.

Economic message: Lehman did suffer a run-like funding collapse, but it occurred mostly through quantities, not sharply rising margins. This is the paper's main dealer-level run evidence.

Table VI

Lehman's Last Days

statistics describing Lehman's tri-party repo position leading up to its bankruptcy. "Margin Spread" is the difference in perc average margin Lehman faced minus the average margin faced by a group of other large dealers. "Share of Nongovernment lateral posted that is classified as nongovernment. Nongovernment securities are those not backed by the full faith and c nent. Examples are equities, corporate bonds, and other securities not issued by the U.S. Treasury or other government ag total number of Lehman's investors in the tri-party repo market. The numbers for July and August are the average ove

			September								
	July	August	2	3	4	5	8	9	10	11	12
	−0.05	−0.13	−0.08	−0.07	−0.10	−0.03	0.03	−0.03	0.48	0.66	1.11
nment	25	28	31	31	31	31	31	32	31	32	39
	70	69	68	68	69	69	69	63	50	48	41

Table VI: Lehman's last days

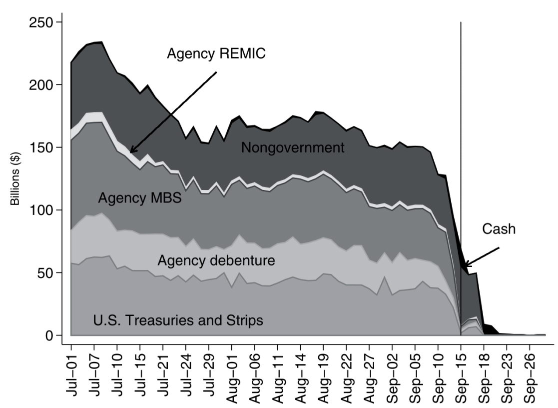

Figure 6. Collateral posted by Lehman Brothers in tri-party repo. This figure presen the composition of collateral posted in tri-party repo by Lehman leading up to its bankrupt The vertical line represents Lehman Brothers' bankruptcy. MBS is mortgage-backed securiti and REMIC is real estate mortgage investment conduits. Nongovernment securities are those n backed by the full faith and credit of the federal government. Although they rarely exercise dealers have the option to post cash as collateral in tri-party repo.

Figure 6: collateral posted by Lehman

6.7 Table VII and Figure 7: CDS regressions and PDCF backstop

Read:

- Table VII relates dealer CDS spreads to margins and collateral values.
- CDS spreads have little explanatory power for margins or collateral amounts in tri-party repo.
- This suggests that tri-party cash investors did not smoothly adjust margins or quantities with measured dealer credit risk.

- Figure 7 shows securities dealers' credit from the Federal Reserve's balance sheet, including spikes around Bear Stearns and Lehman.
- The PDCF likely helped stabilize tri-party repo funding by providing a liquidity backstop.

Economic message: Dealer credit risk did not translate smoothly into higher margins. Public liquidity support may have helped prevent tri-party funding stress from becoming systemic.

Table VII

Estimated Coefficients of CDS Regressions

This table presents estimated coefficients from regressions that relate CDS spread to margins and, separately, value of collateral posted in tri-party repo. Each regression is estimated including only government collateral and then only nongovernment collateral. Both left-hand-side variables (margins and collateral value) and the CDS spread are in log terms. Not reported are the fixed effect coefficients for collateral class, dealers, and day. Std Err is standard error, clustered by dealer. The government regressions had 18,158 observations and the nongovernment regressions had 20,286. Government securities are those backed by the full faith and credit of the federal government.

	Government Collateral		Nongovernment Collateral	
	Estimate	Std Err	Estimate	Std Err
Independent variable: *Margins*				
CDS Spread	0.003	0.001	0.012	0.007
CDS Spread Squared × 100	0.0047	0.0015	0.023	0.00013
R^2	0.36		0.49	
Independent variable: *Collateral value*				
CDS Spread	0.051	0.145	−0.170	0.147
CDS Spread Squared	0.001	0.003	−0.003	0.003
R^2	0.81		0.52	

Table VII: CDS regressions

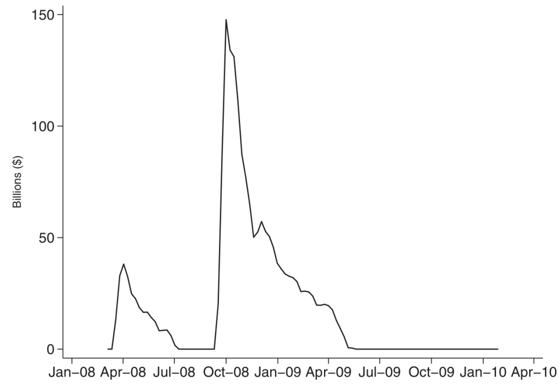

Figure 7. Weekly average of securities dealers' credit. This figure presents a time series securities dealers' credit on the Federal Reserve's balance sheet. Source: Federal Reserve H.4.1

While our paper does not speak directly to whether minimum margins a desirable, it does suggest that the benefits from minimum margins are unlike

Figure 7: securities dealers' credit at the Fed

6.8 Appendix evidence: margin distributions and decomposition

Read:

- Figure C1 shows dispersion in margins across dealers within asset classes.
- Collateral classes ordered as lower quality have higher margin distributions.
- Table C.I decomposes margins into dealer fixed effects and collateral-type fixed effects.
- Dealer fixed effects vary widely, indicating persistent dealer-level differences in margin terms.
- Collateral-type effects rise from agency/Treasury-like collateral to whole loans and private-label CMO collateral.

Economic message: Margins differ by both collateral quality and dealer identity. This reinforces the idea that tri-party repo pricing was cross-sectionally differentiated but not systemically repriced upward in crisis.

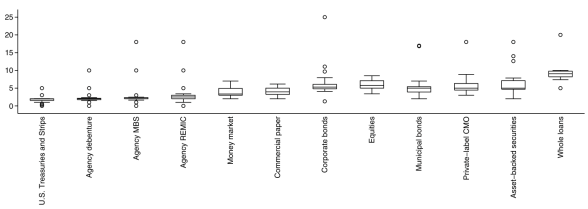

Figure C1: distribution of margins by asset class

Table C.I

Coefficients on Margin Decomposition

This table presents estimated coefficients that describe the differences in margins across dealers. The variable *clrbnk* is a dummy variable equal to 1 for trades settled by one of the clearing banks. The dealer with the lowest average margins and U.S. Treasuries and strips were the excluded dummies for dealer and asset classes, respectively. There are 85,256 observations and the R^2 for the regression is 0.31. Std Err is standard error, clustered by dealer-asset class pair. MBS is mortgage-backed securities, REMIC is real estate mortgage investment conduits, and CMO is collateralized mortgage obligations.

	Coefficient			Coefficient	
Variable	Estimate	Std Err	Variable	Estimate	Std Err
dealer 1	1.07	0.45	Agency Debenture	0.10	0.49
dealer 2	1.32	0.68	Agency MBS	0.57	0.43
dealer 3	1.34	0.81	Agency REMIC	1.09	0.48
dealer 4	1.35	0.68	Money Market	2.16	0.47
dealer 5	1.40	0.34	Other	2.18	0.86
dealer 6	1.44	0.41	Commercial Paper	2.44	0.45
dealer 7	1.45	0.69	Municipal Bonds	3.82	0.54
dealer 8	1.48	0.41	Corporate Bonds	4.01	0.51
dealer 9	1.50	0.36	Equity	4.38	0.70
dealer 10	1.57	0.39	Asset-Backed Securities	4.99	1.02
dealer 11	1.57	0.66	Private-Label CMO	5.37	1.40
dealer 12	1.63	0.25	Whole Loans	6.85	0.78
dealer 13	1.81	0.30			
dealer 14	1.94	0.67			
dealer 15	2.03	0.61			
dealer 16	2.10	0.58			
dealer 17	2.11	0.29	constant	−0.924	0.56
dealer 18	2.16	0.41	clrbnk	0.55	0.45
dealer 19	2.30	0.32			
dealer 20	2.46	0.51			
dealer 21	2.78	0.60			
dealer 22	2.83	0.71			
dealer 23	2.84	0.59			
dealer 24	2.84	0.54			
dealer 25	2.89	0.65			
dealer 26	3.03	0.91			
dealer 27	3.78	1.36			
dealer 28	3.83	0.76			
dealer 29	9.04	1.95			
dealer 30	9.87	4.02			
dealer 31	14.77	2.43			

The omitted dealer has the lowest estimated fixed effect, while the omitted

Table C.I: margin decomposition

7 Short summary

- The paper studies whether the tri-party repo market experienced a system-wide run during the financial crisis.
- It uses confidential daily data from July 2008 to January 2010.
- Unlike bilateral repo, tri-party repo margins were stable for most collateral classes.
- Government collateral, which made up most of the market, had margins that barely moved.
- Nongovernment collateral quantities declined, especially ABS and corporate bonds, but not in a sudden market-wide collapse.
- Dealer-investor relationships were highly stable on a day-to-day basis.
- Lehman Brothers is the main exception: its tri-party repo funding collapsed sharply around bankruptcy.
- Lehman's collapse occurred through falling quantities rather than large margin increases.
- CDS spreads do not explain much of the variation in margins or collateral quantities.
- The PDCF likely helped stabilize the tri-party repo market.
- The policy implication is that repo-market regulation should distinguish tri-party from bilateral repo and margin channels from quantity channels.

8 Conclusion

8.1 Core conclusion

Copeland, Martin, and Walker show that there was no system-wide run on tri-party repo during the financial crisis. The broad tri-party market did not experience the sharp rise in margins documented in bilateral repo. Instead, margins were mostly stable and quantities adjusted gradually, with Lehman Brothers as the key concentrated failure.

8.2 Economic intuition

Tri-party repo cash investors appear to respond to stress differently from bilateral repo lenders. Rather than continuously increasing margins in response to perceived risk, they may continue to roll positions under normal conditions and then withdraw funding sharply when a dealer becomes unacceptable. This makes tri-party repo fragility more nonlinear and dealer-specific.

8.3 Incremental contribution takeaway

The paper's incremental contribution is to replace a one-size-fits-all repo-run narrative with a segmented view of repo markets. Bilateral repo can exhibit margin spirals; tri-party repo can instead show stable margins, stable rollover relationships, and abrupt quantity withdrawal for a failing dealer. This distinction changes how we interpret crisis evidence and how we design repo-market policy.

8.4 Broader takeaway

The paper teaches that shadow banking fragility cannot be diagnosed from haircuts alone. Quantities, collateral composition, market segment, clearing-bank infrastructure, and public liquidity backstops all matter. A complete theory of repo runs must therefore combine margin dynamics with funding-withdrawal dynamics and market plumbing.